Question 1

Determine the reactions (vertical and horizontal) at supports **A** and **B** and the forces (vertical and horizontal) exerted by the pin at C on member BCD of the frame shown in Figure 1.

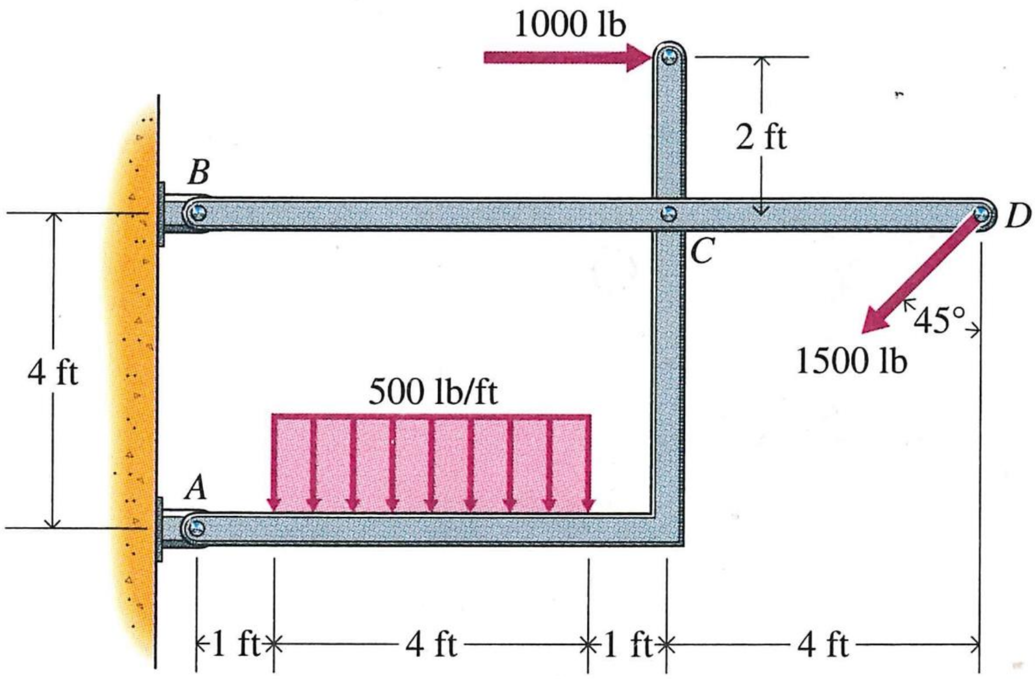

Figure 1

Answers:

Ax = 4651.65 lbs

Ay = 3767.77 lbs

Bx = -4590.99 lbs

By = -707.11 lbs

Cx = -5651.65 lbs

Cy = -1767.77 lbs

Question 2

Three bars are connected with pin joints a shown in Figure 2. The weight of the bars can be neglected. Note that the downward force 500N is acting at the mid-point of member BD and on the pin at C.

2.1. Determine all reactions at E and A.

2.2. Hence, determine all forces at B, C and D.

Note:
Show all free-body diagrams necessary and workings.
All dimension are in meters (m)

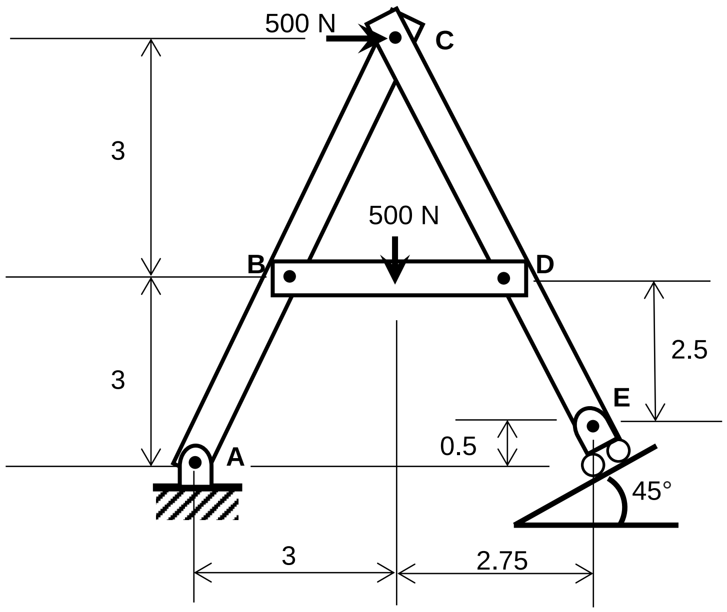

Figure 2

Answers:
Ex = 720 N
Ey = 720 N
Ax = 220 N
Ay = -220 N
Bx = -785 N
By = 250 N
Cx = 65 N
Cy = 470 N
Dx = -785 N
Dy = 250 N